

DANG TRAN NAM

Software Engineer Intern · Hoa Hung Ward, Ho Chi Minh

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A detail-oriented Software Engineer with a strong interest in backend/database architecture and performance optimization. I strive to build reliable, high-efficiency systems by understanding the fundamental interactions between software and hardware. Committed to writing clean, maintainable code and continuously optimizing system latency through data-driven approaches.

Experience

An Binh Hospital

Software Engineer Intern – IT Asset Management (ITAM): Distributed Telemetry System

- **Distributed Telemetry System:** Designed and developed a cross-platform agent in Go to harvest real-time CPU, RAM, and Disk metrics from over 100+ edge devices, providing centralized monitoring.
- **Resilience & Reliability:** Implemented Circuit Breaker pattern with Exponential Backoff and Jitter logic, preventing Thundering Herd issues during agent reconnection and ensuring 99.9% uptime for the monitoring system.
- **Performance Optimization:** Reduced memory footprint of the agent by over 60% (to under 15MB) by implementing efficient log rotation and strictly typed data structures.
- **Security:** Secured communications using Bearer Token authentication and SHA-256 HMAC signatures, ensuring data integrity and access control.

Projects

## High-Concurrency Inventory Engine

**Problem:** Massive transaction volume during peak traffic caused race conditions risking over-selling and inventory bugs.

**Solution:** Optimistic Locking with version columns to resolve race conditions, ACID transactions to guarantee write-side data integrity, and multi-column indexes to avoid full table scans.

**Result:** Average query time down 50%, zero over-selling under concurrent writes.

**Tech Stack:** Go, PostgreSQL, Raw SQL for fine-grained performance control.

### Skills

**Hard Skills:** Java, Golang, C/C++, Linux

### Education

**Saigon University (SGU)** – Software Engineering (2021 - 2026)  
Engineer Degree

### Certifications

- TOEIC (760 R/L)
- AWS Cloud (In Progress)